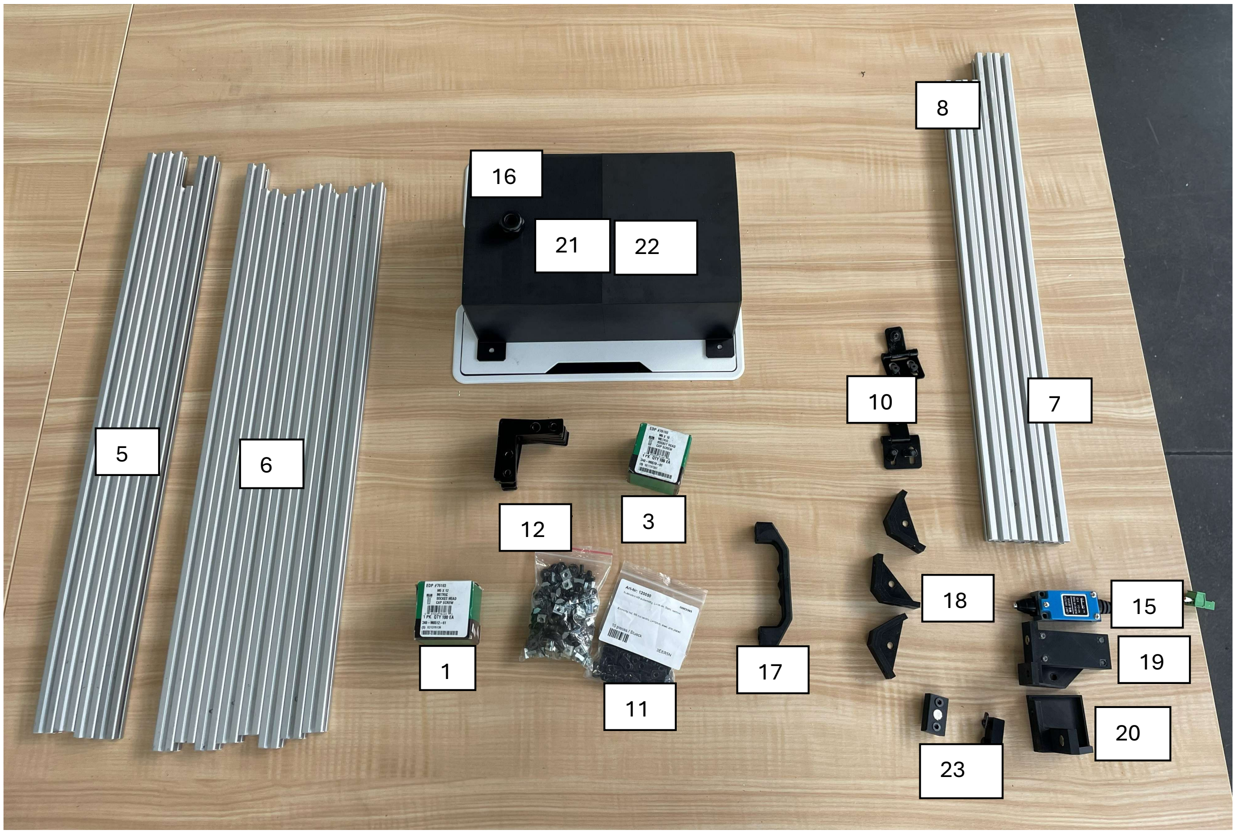
Desktop CNC Mill Enclosure Instructions



FIND ALL OF THESE ITEM DESCRIPTIONS IN “LC-CNC-BoM” FOR ITEM LINKS

	Description	Qty
1	M5x12 mm bolts (100 pack)	1
2*	M5 hex nut (100 pack)	1
3	M3x10mm bolts (100 pack)	1
4*	Magnets (30 pack)	30
5	20-2020 LENGTH: 596.6 millimeters	8
6	20-2020 LENGTH: 636.6 millimeters	4
7	20-2020 LENGTH: 593.6 millimeters	2
8	20-2020 LENGTH: 553.6 millimeters	2
9*	14157 Economy Gasket LENGTH: 8 meters	1
10	Hinges	2
11	T-Nuts (10 pack)	10
12	L Connector Bracket (10 pack)	2
13*	Polycarbonate sheet (24x24x1/8”)	2
14*	Loctite-Blue	1
15	Limit Switches	1
16	Cord grip	
	3D PRINTS	
17	CNC Mill Door Handle.STEP	1
18	Corner Bracket for Hinge Door.STEP	3
19	Corner Bracket with Magnet Bottom V2.STEP	1
20	Corner Bracket with Magnet Top.STEP	1
21	Enclosure cover 1 v1.step	1
22	Enclosure cover 2 v1.step	1
23	Magnet Frame Mount.STEP	2
24*	Connector cover bottom.STEP	7
25*	Connector cover top.STEP	7
	Custom Polycarbonate Sheets (IF NOT SENT THESE, ORDER FIVE 24x24x1/8” SHEETS AND CUT TO SPECIFICATION)	
26*	Polycarbonate top panel.dxf (File in 3D printing files>Enclosure, then custom cut 24x24x1/8”)	1
27*	Polycarbonate back panel (Refer to step 7, drill two 1” holes into lower corners of 24x24x1/8”)	1
28*	Polycarbonate door panel (Trim polycarb sheet to 22.2x22.2x1/8”)	1

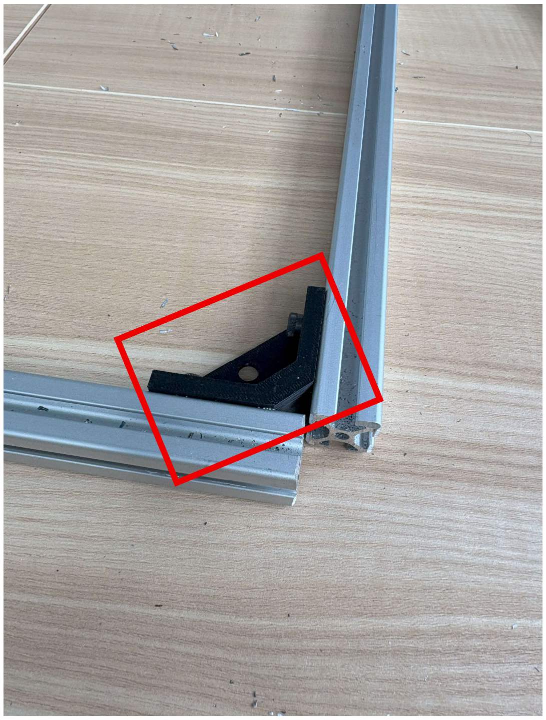
**Not referenced*

1. Gather M5*12 bolts and t-nuts, then attach these to the corner brackets. Make sure these bolts are loosely tightened.

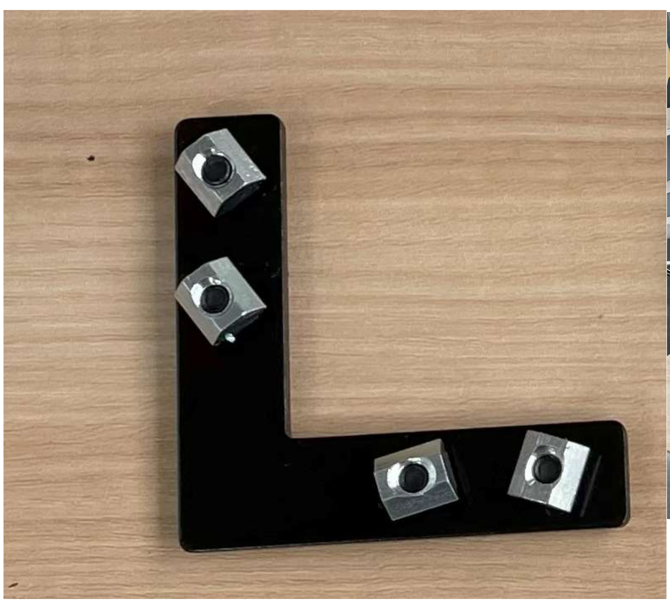

2. Gather your magnets, magnet frame mount, bottom corner bracket, and top corner bracket. Apply super glue to the magnets and holes as shown. Make sure the magnet frame mount attracts to both top and bottom corner brackets.

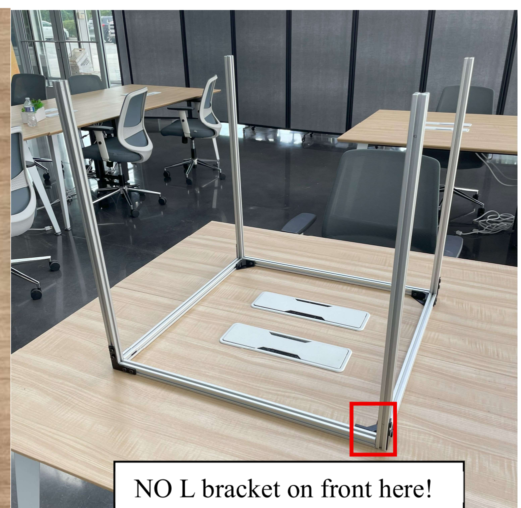


3. Take the 596mm extrusions and slide them into the corner brackets as such.

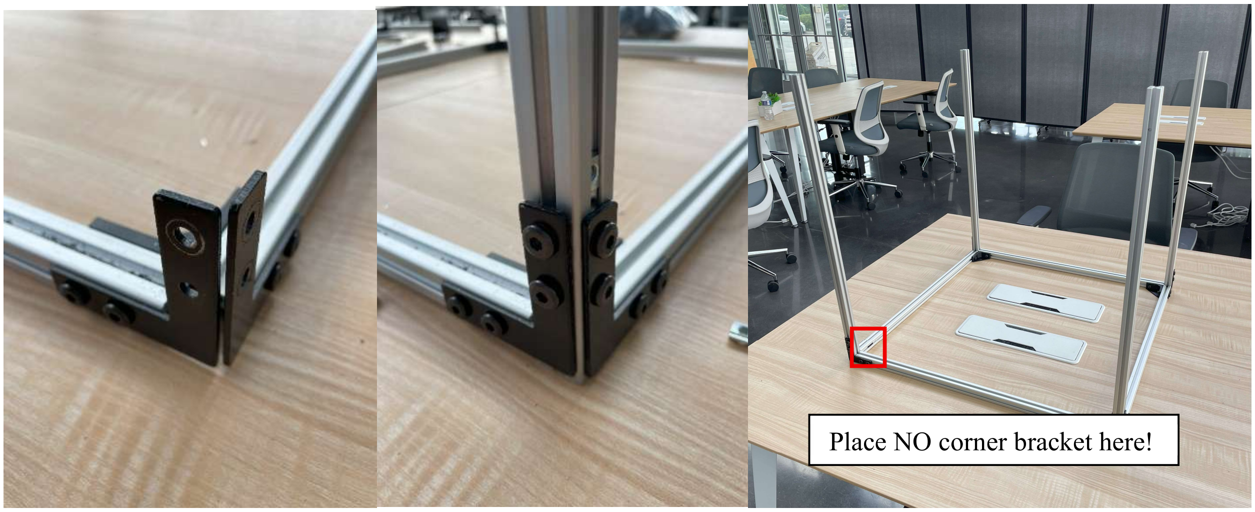


4. Gather the appropriate M5 bolts and T-nuts that came with the aluminum L brackets and attach them to the 636mm 8020 extrusions as such. With the only empty channel being the one where the door hinges are placed. Again, make sure the bolts are loosely tightened.

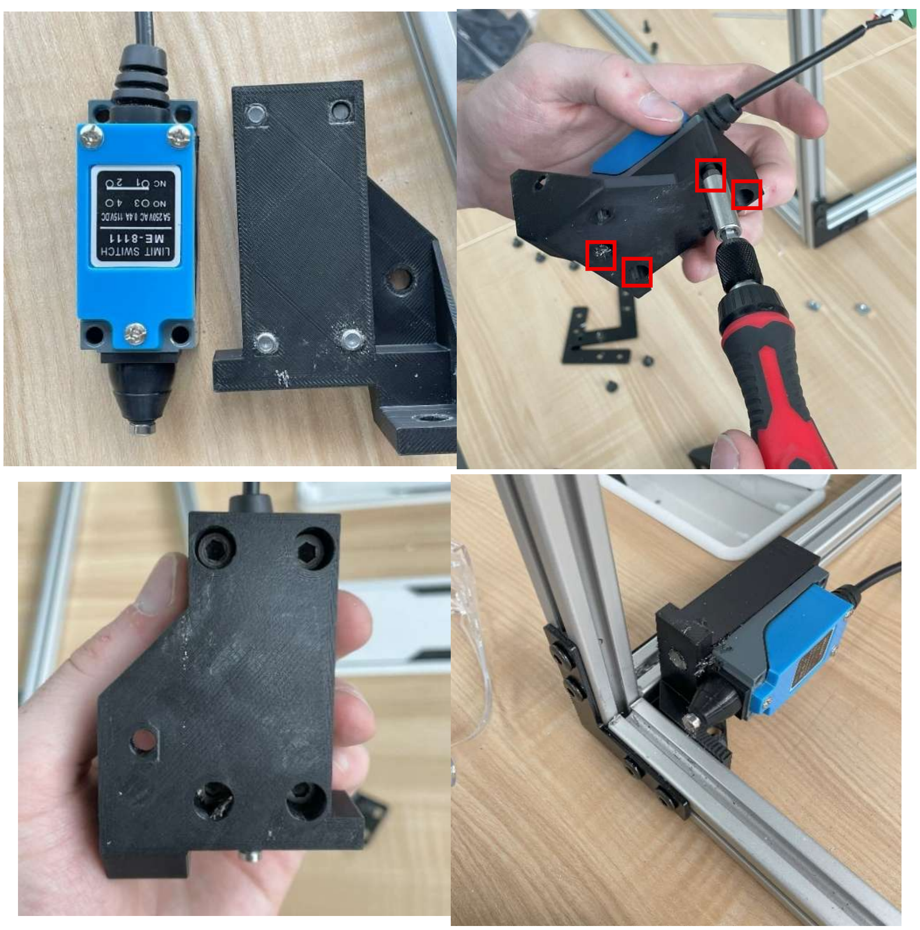




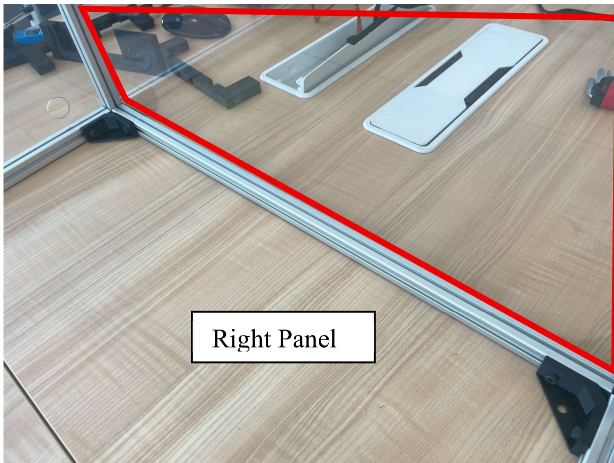
5. Place the 636mm 8020s in between the 596mm ones as shown. Adjust the position of the 8020 extrusions as needed. Then tighten all the bolts once the desired position is reached. Repeat for all four corners. Notice that one corner does not have a corner bracket, this will be addressed in the next steps.

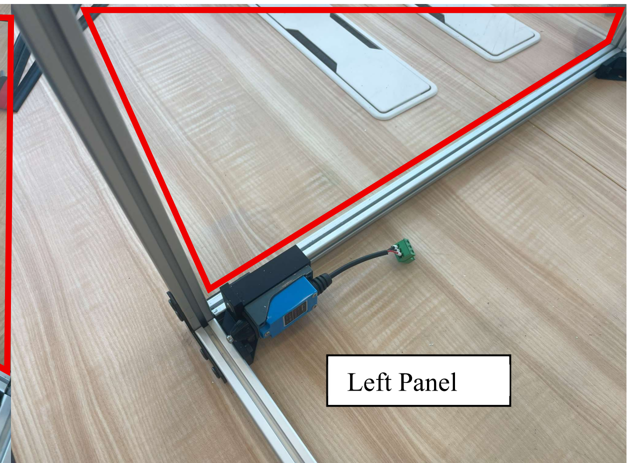


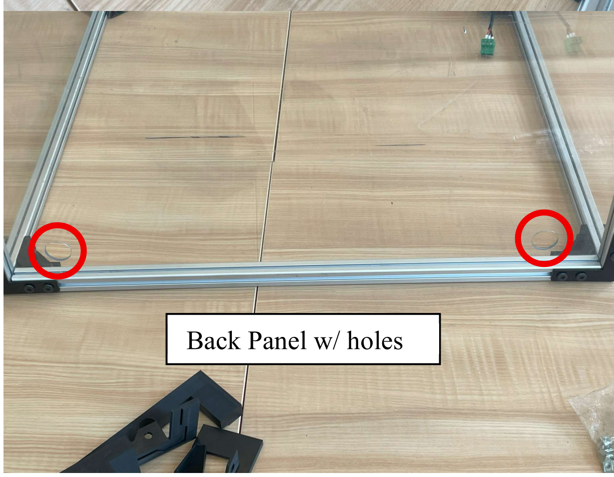
6. For the corner bracket that holds the limit switch, find the M5*12 bolts and T-nuts then screw down the limit switch as shown.



7. Next, slide in all the polycarbonate sheets as shown, besides the side where the door should be. Also, make sure that the polycarbonate sheet with the holes in the bottom corners is the one that is placed on the back side of the machine.

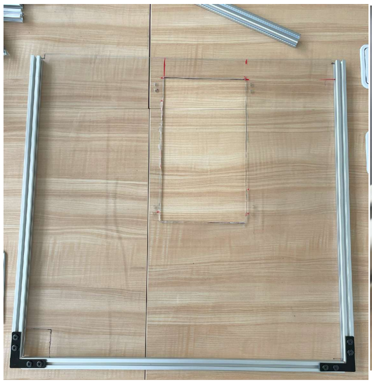




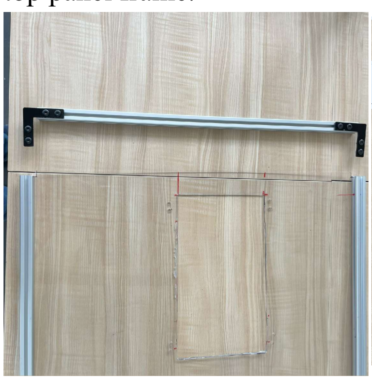


8. With the panels in place, prepare aluminum L brackets with its M5 bolts and T-nuts that came with them and attach them to all four corners as such.

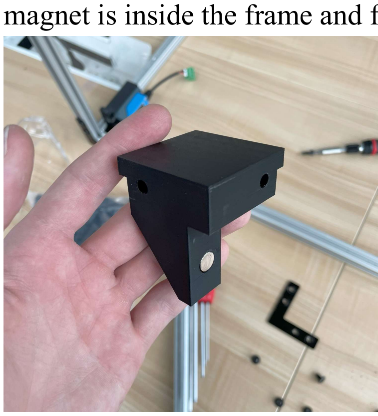
9. With the L brackets in place on each corner, grab the custom cut polycarbonate top panel, four 596mm extrusions, and four L brackets with its corresponding M5 bolts and nuts. Connect three extrusions together with two L brackets and slide in the top panel to keep the corners square. Tighten bolts. Slide in two T nuts into the large cutout corner. Make sure the cutout corner is in the bottom left corner.

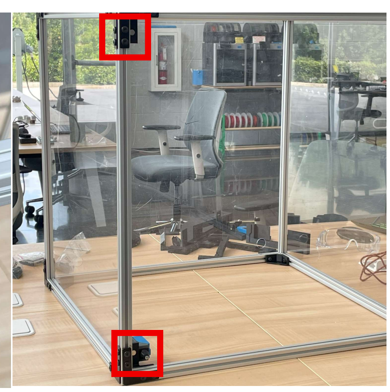



10. Attach the last 596mm extrusion as shown. Tighten bolts. Insert four T nuts, two on each end of all four extrusions, oriented where the L brackets on each corner of the frame can grab onto the top panel frame.



11. With T nuts in place, slide entire top panel frame into each corner of the frame and mount with corresponding M5 bolts into the inserted T nuts. Ensure the large corner cutout is above the safety switch. Tighten bolts. Place the top corner bracket into the large corner cutout and ensure the magnet is inside the frame and facing the same direction as the limit switch underneath it.

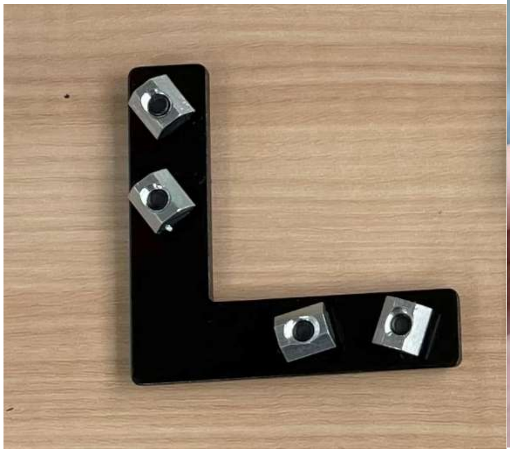




12. Use three M3 bolts to attach the two Z carriage enclosure parts together. Orient the Z carriage enclosure as shown and use four M5*12 bolts and respective nuts to attach to the top panel. Attach the cable gland as shown.

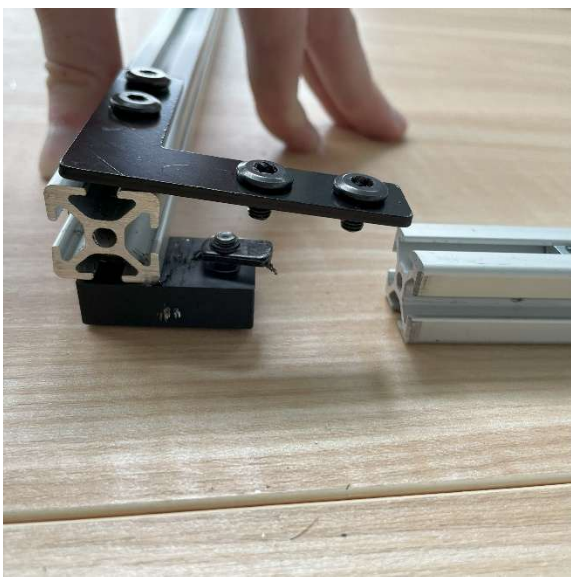
13. Now, for the door, gather the proper extrusions (553mm and 593mm), M5 bolts, T-Nuts, L brackets, magnets, hinge and door handle.

14. Prepare the magnets, L brackets and door handle attaching the proper bolts and T-nuts as shown.



15. Slide the door handle into the left 553mm extrusion as shown, then tighten it down to the middle of that extrusion.

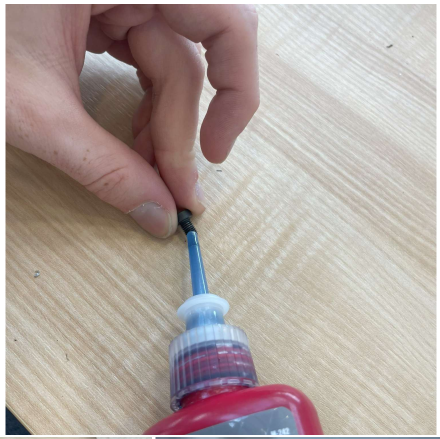

16. Then take your magnets and L brackets, attach them to the 553mm 8020 with the door handle and loosely secure these as shown.

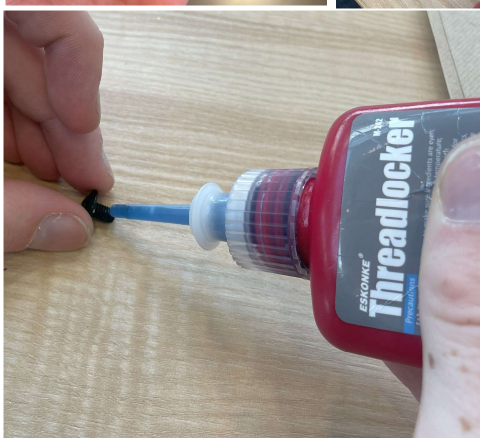


17. Now slide your 593mm extrusion in, with the 553mm extrusion fitted in between the two as shown and tighten down your brackets and magnets.

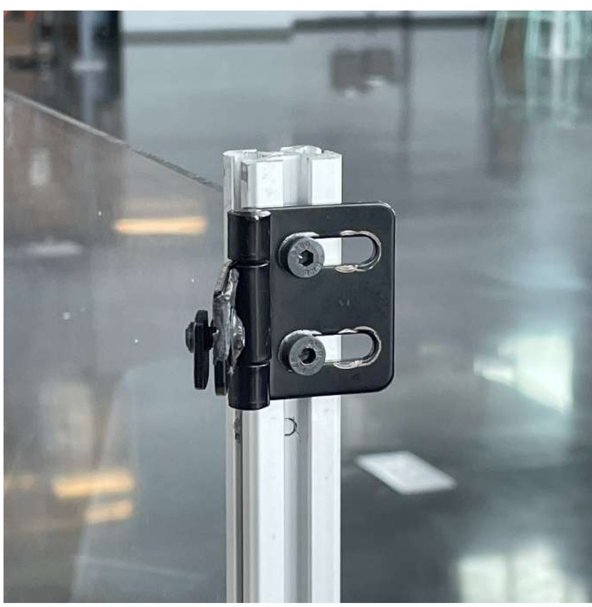
18. Slide in the front panel 22.2x22.2x1/8” polycarbonate sheet and attach the final L brackets to the opposite end of your door frame with the brackets facing inside the enclosure as shown.

19. Before attaching the door hinges onto the right extrusion, make sure to apply thread locker to each bolt that came with the door hinges, then loosely attach T nuts onto each bolt.





20. Now, take the door hinges with the appropriate bolts and nuts attached, and slide them from the bottom of the right 636mm extrusion on enclosure frame.

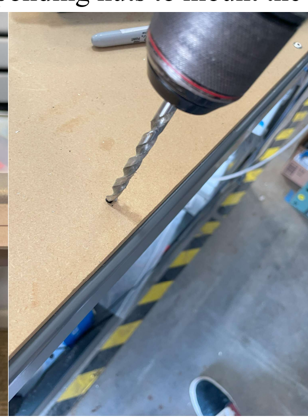


21. Adjust the position of the hinges to bias the position of the door towards the top of the enclosure, leaving a larger gap at the bottom. When you tighten the bolts for the hinges down, the gaps at the top and bottom should be a similar distance.

22. Finally, tighten the hinges to the door as shown. (when testing the door hinge, it may scrape the bottom extrusion, this will most likely be solved when all four corners of the enclosure are mounted on the table.)

23. Take the enclosure and place on your table where it should be located with respect to your machine. Mark all four mounting holes on the table where we will need to drill through.

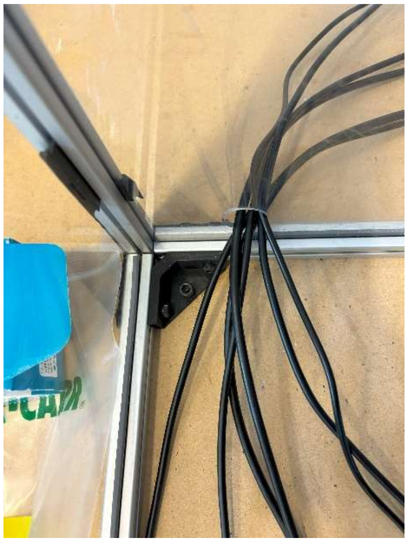

24. Drill through the table with a H 0.266” drill bit and find four 1/4-20 bolts of appropriate length for your table and corresponding nuts to mount the enclosure on the table.



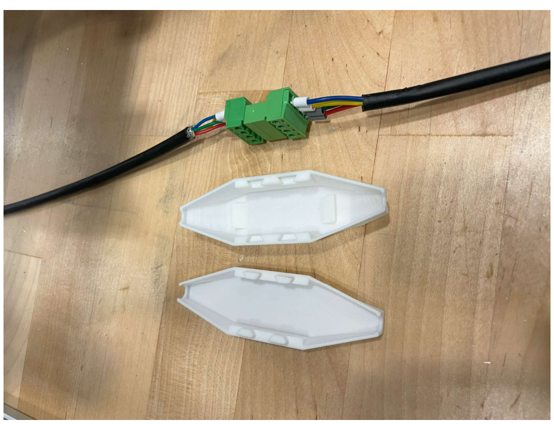
25. With milling machine in its desired location, place the enclosure over the machine and use a 3/16" Allen wrench to tighten the bolts and corresponding nuts.


26. Route your spindle wire through the top of the enclosure as shown. Make sure to put on the cap over the wire to properly hold the cable.


27. Route all wires for the milling machine through the holes for in the rear panel.



28. If you find the door still scrapes the bottom extrusion for the enclosure as it opens and closes, readjusting the hinges to bias the door towards the top of the enclosure should give enough room for the door to open and close smoothly. Make sure to reapply thread locker to the hinge bolts after readjustment.
29. If the electrical box is built and wires have been run into the enclosure, connector covers can be used to protect connectors from loose chips and other debris. Simply place the connection in one half of a connector cover pair, place the other half over the top and press until the plastic snaps into place.




This cover can be undone by separating the halves.